F12 4th CAR

Lecture 10, part I: Section 1.6 A Library of Functions

Linear function

Square function

Cube function

Square root function

Reciprocal function

Absolute value function

Piece-wise defined function

Lecture 10, part II: Section 1.7 Transformations of Functions

Vertical and horizontal translations

Reflections across an axis

Vertical and horizontal compressions and stretches

Rigid and nonrigid transformations

Reuse 1-95

Lecture 10, part I: Section 1.6 A Library of Functions

slope

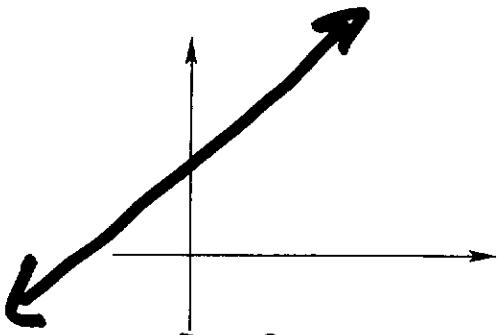
1. Linear Function $f(x) = mx + b$

y-intercept

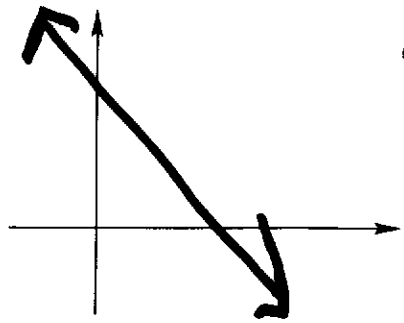
For both

D: $(-\infty, \infty)$

R: $(-\infty, \infty)$



$m > 0$

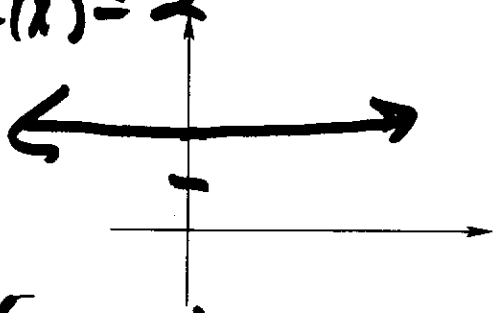


$m < 0$

NOTE: There are two special types of linear functions: **Constant Function**, $f(x) = c$ and **Identity Function**, $f(x) = x$. or $y = x$

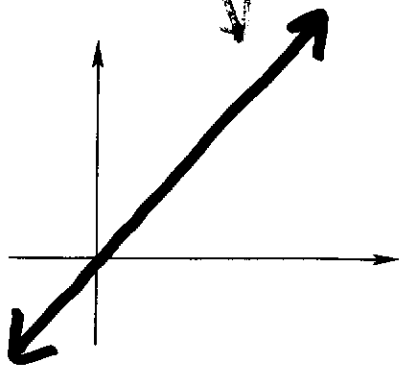
$y = 2$

$f(x) = 2$



D: $(-\infty, \infty)$

R: $\{2\}$ $m = 0$



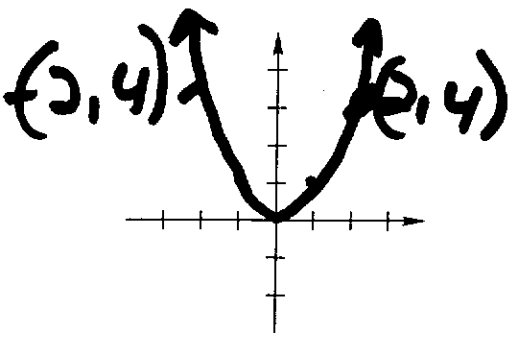
D: $(-\infty, \infty)$

R: $(-\infty, \infty)$

$m = 1$

$y = 1x + 0$

2. Square Function $f(x) = x^2$



D: $(-\infty, \infty)$

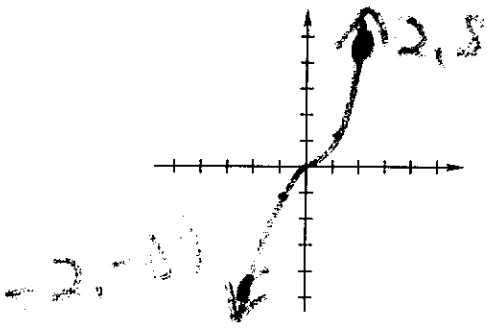
R: $[0, \infty)$

sym wrt y-axis

even function

$(x,y) \to (-x,y)$

3. Cube Function $f(x) = x^3$



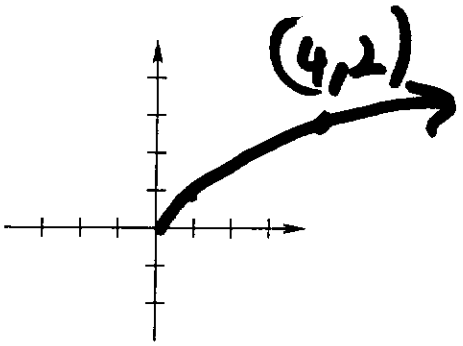
D: $(-\infty, \infty)$

R: $(-\infty, \infty)$

sym wrt to the origin

odd [illegible]

4. Square Root Function $f(x) = \sqrt{x}$



D: $[0, \infty)$

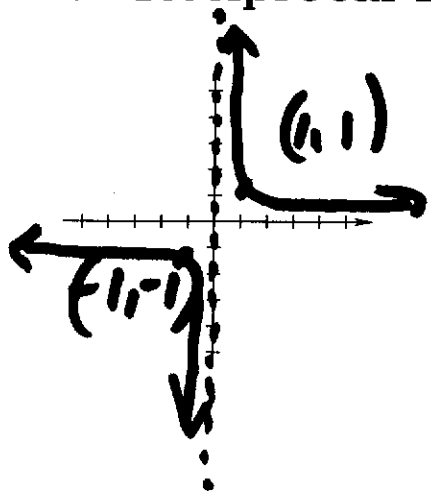
R: $[0, \infty)$

no sym.

D: $(-\infty,0)\cup(0,\infty)$

R: $(-\infty,0)\cup(0,\infty)$

5. **Reciprocal Function** $f(x) = \dfrac{1}{x}$



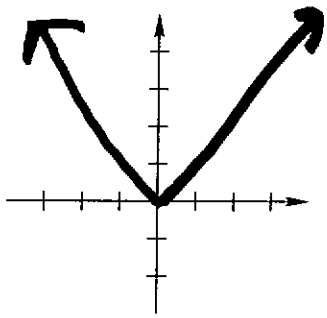
sym wrt origin

so odd funct.

$f(-x) = -f(x)$

$(x,y) \to (-x,-y)$

6. **Absolute Value Function** $f(x) = |x|$



D: $(-\infty,\infty)$

R: $[0,\infty)$

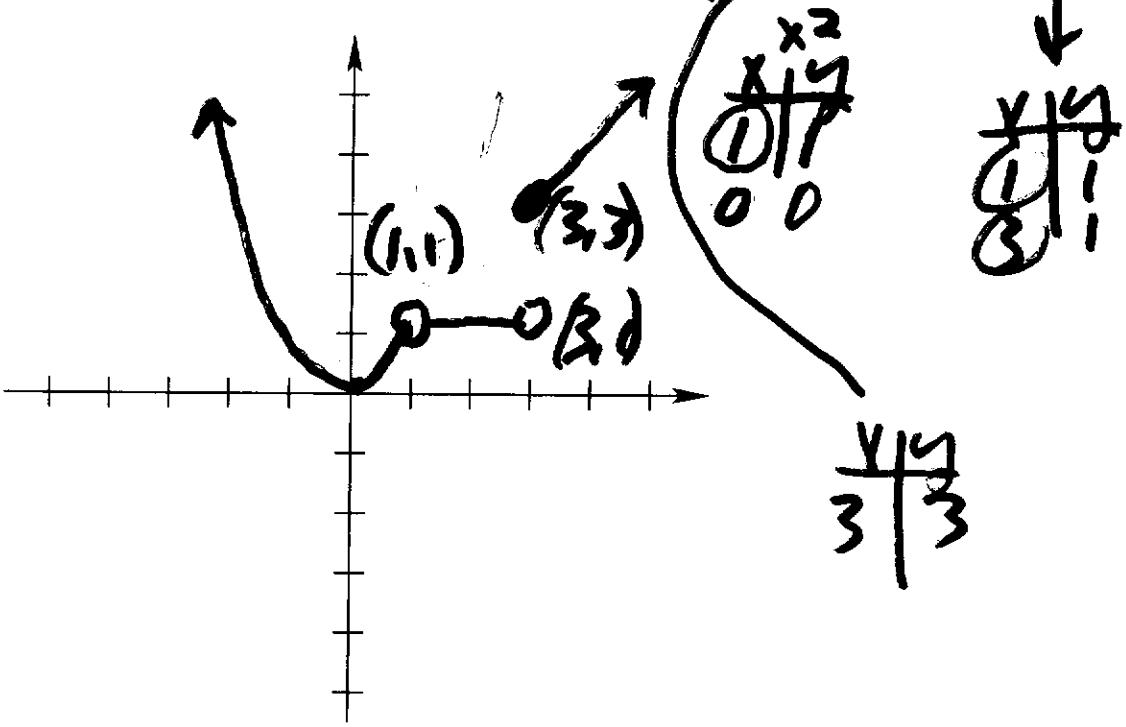
even function

A **piecewise-defined function** is defined by different rules on different parts of its domain.

Graph a piecewise-defined function

1. Divide the domain of the function into parts.
2. Graph separately each branch of the corresponding parts of the domain.
3. On each branch of the graph, draw end points as a solid dot • if the interval is closed on that side and as ∘ if it is open.

ex. Sketch the graph of $f(x) = \begin{cases} x^2, & x < 1 \\ 1, & 1 < x < 3 \\ x, & x \geq 3 \end{cases}$



Lecture 10, part II: Section 1.7
Transformations of Functions

Vertical and Horizontal Translations

Given $y = f(x)$ and $c > 0$:

To Graph:	Shift $f(x)$
vertical shifts: $f(x) + c$	↑ c units
$f(x) - c$	↓ c units
horizontal shifts: $f(x + c)$	← c units
$f(x - c)$	→ c units

NOTE:

Numbers inside parentheses change the **argument** (x-values): **horizontal translation**

Numbers outside parentheses change the **function values** (y-values): **vertical translation**

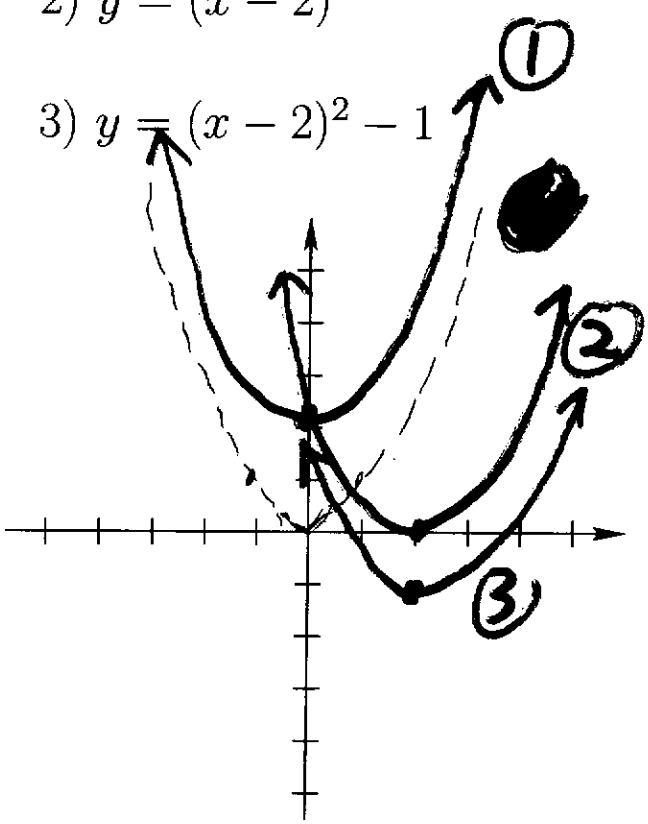
[redacted] $y = x^2$

ex. Use the graph of $y = x^2$ to sketch:

$\uparrow 2$ 1) $y = x^2 + 2$

$\rightarrow 2$ 2) $y = (x-2)^2$

$\rightarrow 2, \downarrow 1$ 3) $y = (x-2)^2 - 1$



Vertical and horizontal shifts generate a family of functions, each with the **same shape** but at **different location** in the xy-plane.

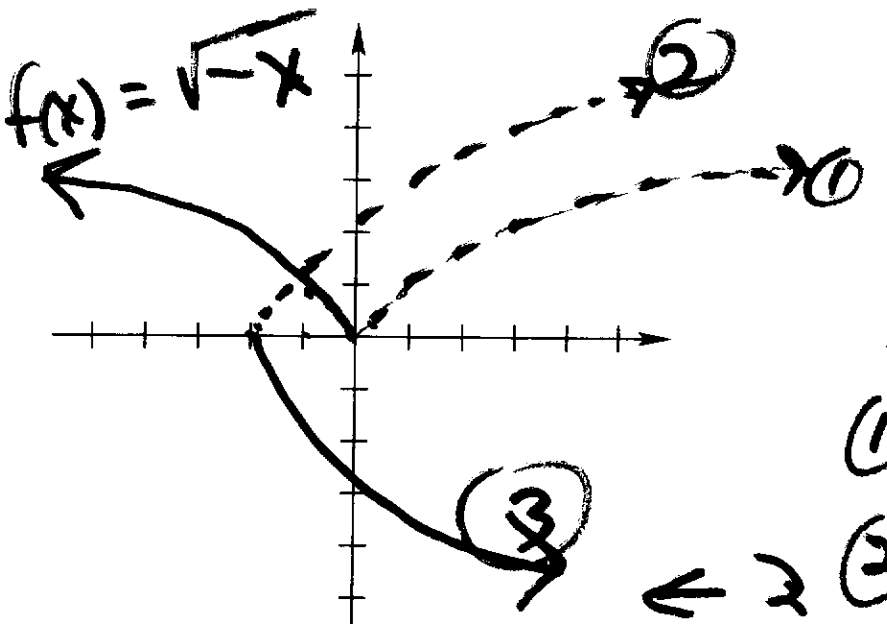
Reflections across an Axis

The graph $y = -f(x)$ is the **reflection about the x-axis** of the graph of $y = f(x)$.

The graph $y = f(-x)$ is the **reflection about the y-axis** of the graph of $y = f(x)$.

NOTE: Do horizontal translations and reflections first. Vertical translations are done last.

<u>**ex.**</u> Sketch $f(x) = \sqrt{-x}$ and $g(x) = -\sqrt{x+2}$ and find the domain of f and g.



$y = \sqrt{x}$

$g(x) = -\sqrt{x+2}$

$\leftarrow 2$

(1) $y = \sqrt{x}$

(2) $y = \sqrt{x+2}$

(3) $y = -\sqrt{x+2}$

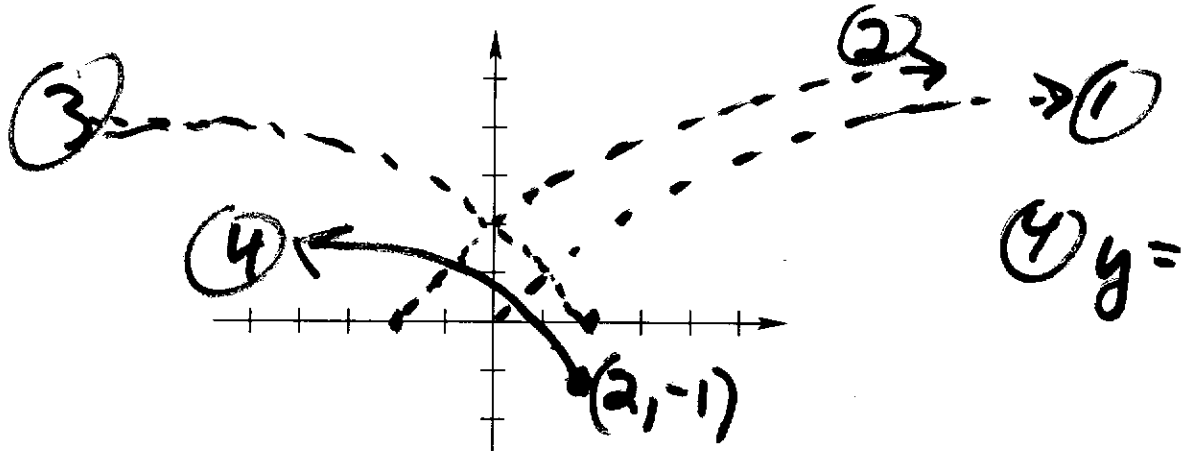
① $y = \sqrt{x}$

② $y = \sqrt{x+2}$ ← 2

③ $y = \sqrt{-x+2}$ refl. about y-axis

ex. Sketch $y = \sqrt{-x+2} - 1$.

Method I: Use $y = \sqrt{x}$.



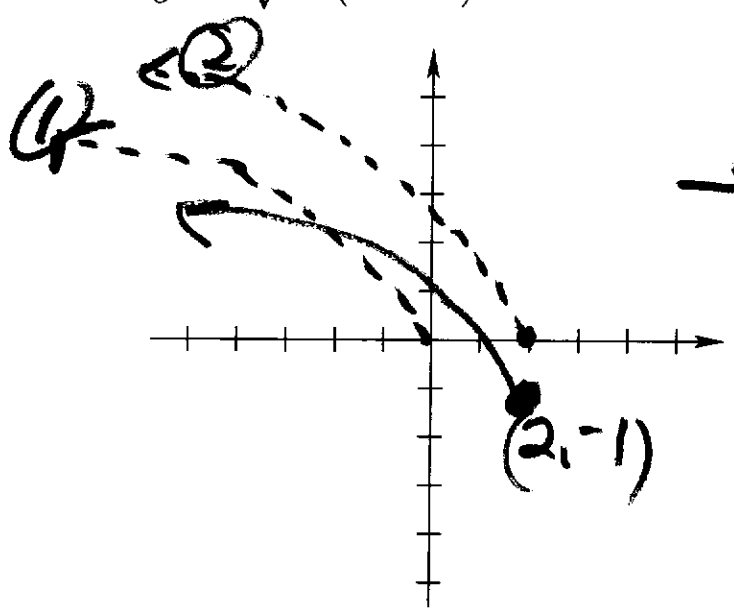
④ $y = \sqrt{-x+2} - 1$

↓ 1

Take $y = \sqrt{-x+2} - 1$ and rewrite by factoring the neg to the front.

$y = \sqrt{-(x-2)} - 1$

Method II: Use $y = \sqrt{-x}$ and write $y = \sqrt{-(x-2)} - 1$.



① $y = \sqrt{-x}$

→ 2 ② $y = \sqrt{-(x-2)}$

↓ 1 ③ $y = \sqrt{-(x-2} - 1$

Vertical and Horizontal Compressions and Stretches

If $c > 1$,

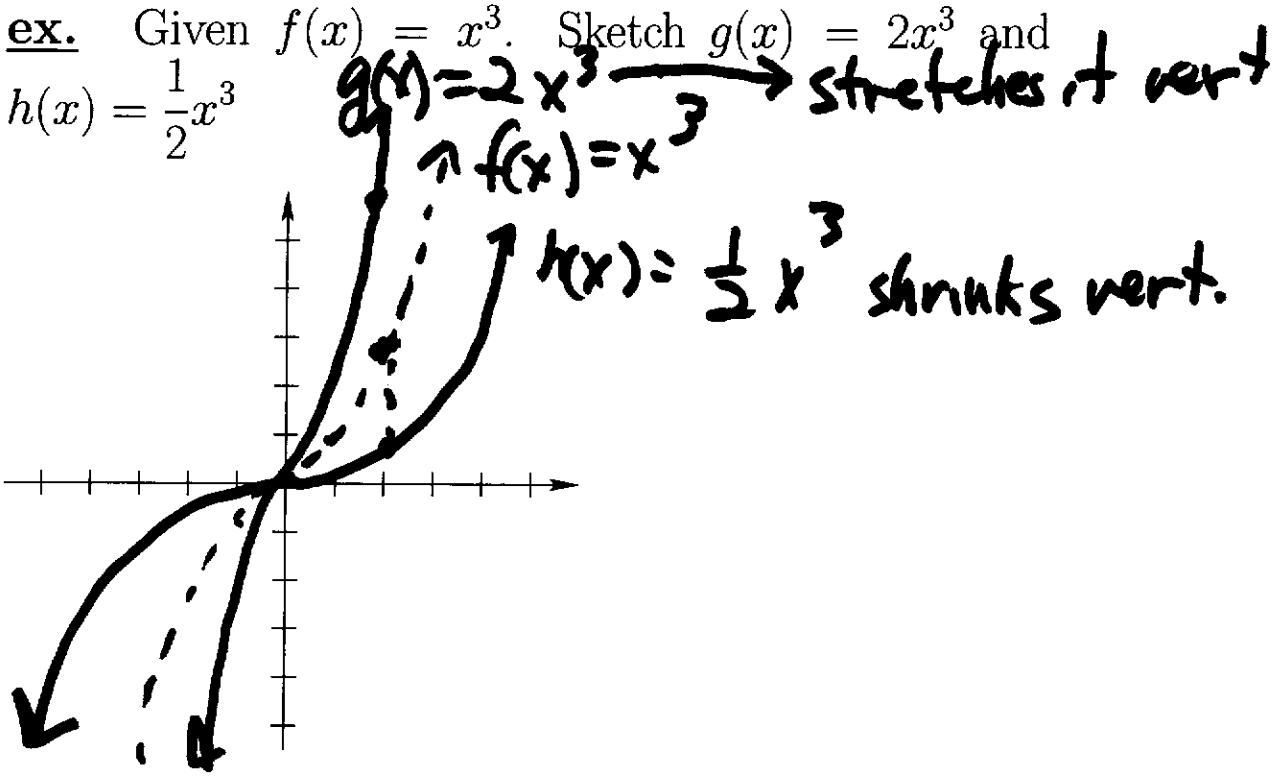
$y = cf(x)$ stretches the graph vertically by a factor of c.

$y = \frac{1}{c}f(x)$ shrinks (or compress) the graph vertically by a factor of c.

$y = f(cx)$ shrink (or compress) the graph horizontally by a factor of c.

$y = f\left(\frac{x}{c}\right)$ stretch the graph horizontally by a factor of c.

ex. Given $f(x) = x^3$. Sketch $g(x) = 2x^3$ and $h(x) = \frac{1}{2}x^3$



Rigid and Nonrigid Transformations

Translations (shifts) and **reflections** are rigid transformation because the basic shape of the graph is unchanged, only the position of the graph in the xy-plane is changed.

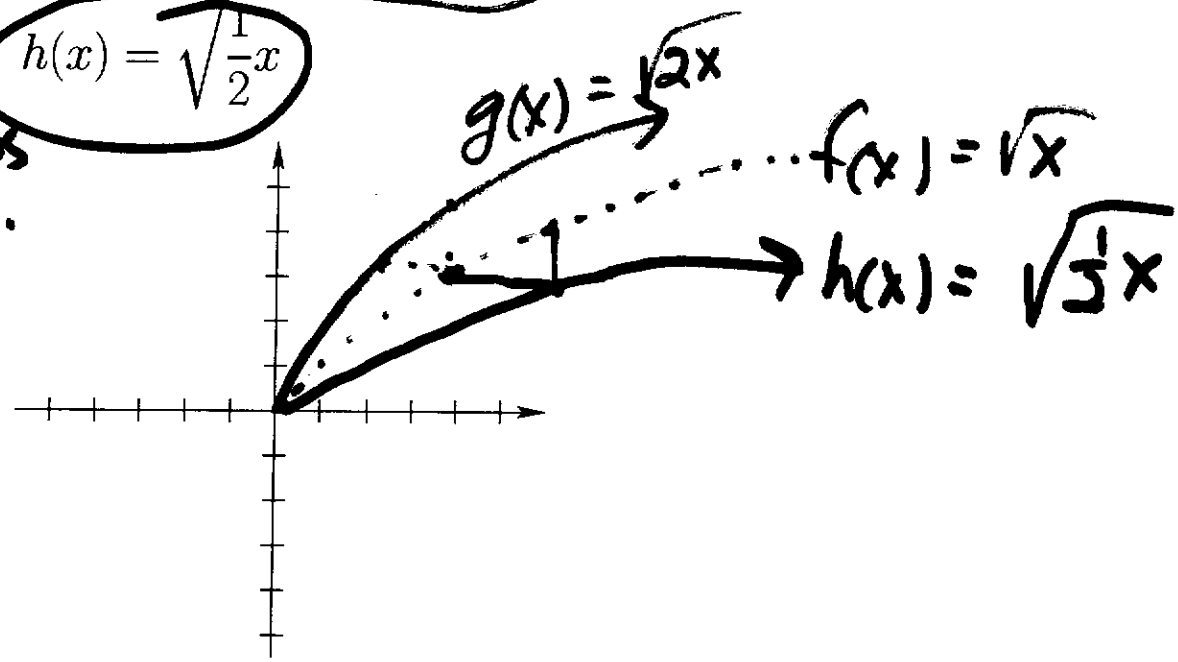
Compression and stretching are nonrigid transformations, they cause a distortion to the shape of the original graph.

Can plot pts to feel more comfortable.

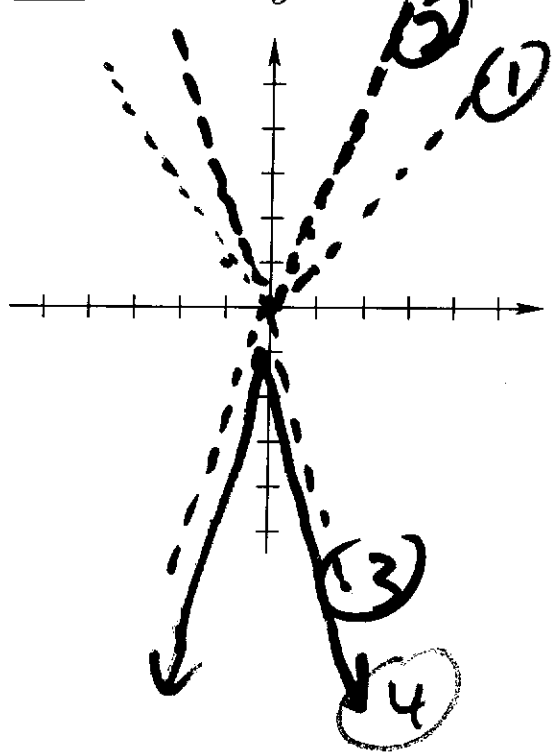
shrink horz

ex. Given $f(x) = \sqrt{x}$. Sketch $g(x) = \sqrt{2x}$ and $h(x) = \sqrt{\frac{1}{2}x}$

stretches horiz.



ex. Sketch $y = -|2x| - 1$



(1) $y = |x|$

(2) $y = |2x|$

shrinks hori. (or stretches vert.)

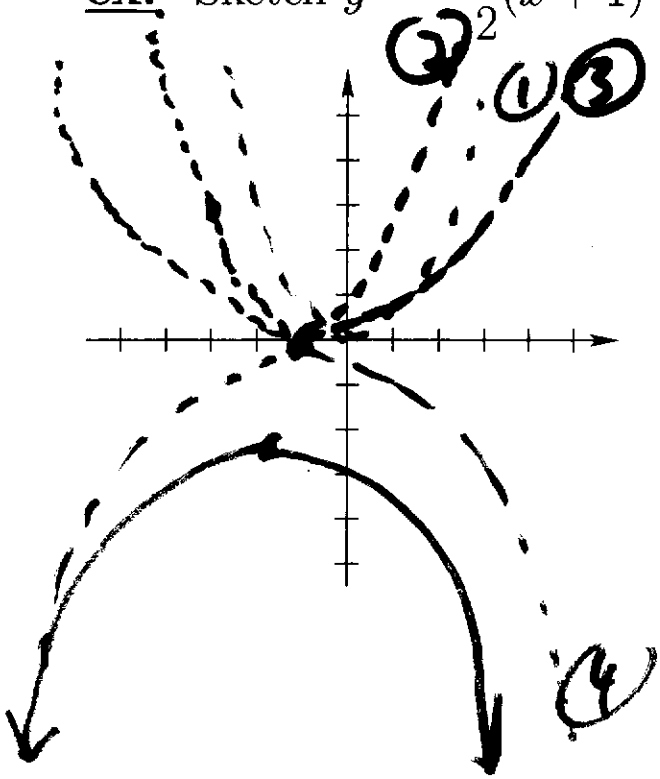
(3) $y = -|2x|$ refl. across x-axis

(4) $y = -|2x| - 1$ ↓1

① $y = x^2$

← 1 ② $y = (x+1)^2$

ex. Sketch $y = -\frac{1}{2}(x+1)^2 - 2$



③ $y = \frac{1}{2}(x+1)^2$

shrinks vert

④ $y = -\frac{1}{2}(x+1)^2$

refl. across x-axis

⑤ $y = -\frac{1}{2}(x+1)^2 - 2$

↓ 2

Practice.

Let $(1, 1)$ be a point on the graph of $y = f(x)$. Find two points on the graph of $y = f(x+2) - 1$,

1) if $y = f(x)$ is an odd function.

2) if $y = f(x)$ is an even function.

Answer. 1) $(-1, 0)$, $(-3, -2)$ 2) $(-1, 0)$, $(-3, 0)$